Mathematics 101

Integrals, Vectors, Sequences, and Trigonometry

Allowed aids: Calculator, writing materials, and formula sheet.

Task 1

Compute the integrals.

a) $\displaystyle\int_0^{\pi} x\cos x\,dx$

Write your calculations/justification in this box

Suggested solution

Integration by parts with $u = x$, $v' = \cos x$:

$$\begin{aligned}
\int_0^{\pi} x\cos x\,dx &= [x\sin x]_0^{\pi} - \int_0^{\pi} \sin x\,dx && (1)\\
&= 0 - [-\cos x]_0^{\pi} && (2)\\
&= -(-\cos\pi + \cos 0) && (3)\\
&= -(1+1) = -2 && (4)
\end{aligned}$$

b) $\displaystyle\int \frac{1}{x^2+x}\,dx$

Write your calculations/justification in this box

Suggested solution

Partial fraction decomposition, $x^2 + x = x(x+1)$:

$$\frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}$$

$$\int \frac{1}{x} - \frac{1}{x+1}\, dx = \ln|x| - \ln|x+1| + C$$

Task 2

Determine t such that

$$\int_1^t \frac{1}{x+1}\, dx = \ln\left(\frac{5}{2}\right)$$

Suggested solution

Integrate the left-hand side using the substitution $u = x+1$, $u' = 1$:

$$\begin{aligned}
\int_1^t \frac{1}{x+1}\, dx &= \int_1^t \frac{1}{u}\frac{1}{u'} du && (5)\\
&= \int_1^t \frac{1}{u} du && (6)\\
&= [\ln|u|]_1^t && (7)\\
&= [\ln|x+1|]_1^t && (8)\\
&= \ln(t+1) - \ln 2 && (9)
\end{aligned}$$

Set this equal to the right-hand side and solve for t:

$$\begin{aligned}
\ln(t+1) - \ln 2 &= \ln\left(\frac{5}{2}\right) && (10)\\
\ln(t+1) &= \ln\left(\frac{5}{2}\right) + \ln 2 = \ln\left(\frac{5}{2}\cdot 2\right) = \ln 5 && (11)\\
t+1 &= 5 && (12)\\
t &= 4 && (13)
\end{aligned}$$

Task 3

The functions f and g are given by $f(x) = 4 - x^2$ and $g(x) = x+2$. The graphs of f and g bound a region. What is the area of this region?

Suggested solution

You should also sketch the graphs in a coordinate system for $x \in [-2, 2]$.
Find the intersection points:

$$
\begin{aligned}
f(x) &= g(x) && (14)\\
4 - x^2 &= x + 2 && (15)\\
0 &= x^2 + x - 2 && (16)\\
&= (x-1)(x+2) && (17)
\end{aligned}
$$

The graphs intersect at $x = -2$ and $x = 1$.
The area is given by the integral

$$
\begin{aligned}
\int_{-2}^{1} (f(x) - g(x))\,dx &= \int_{-2}^{1} (2 - x - x^2)\,dx && (18)\\
&= \left[2x - \frac{x^2}{2} - \frac{x^3}{3}\right]_{-2}^{1} && (19)\\
&= \left(2 - \frac{1}{2} - \frac{1}{3}\right) - \left(-4 - 2 + \frac{8}{3}\right) && (20)\\
&= \frac{7}{6} - \left(-6 + \frac{8}{3}\right) && (21)\\
&= \frac{7}{6} + 6 - \frac{8}{3} && (22)\\
&= \frac{7}{6} + \frac{36}{6} - \frac{16}{6} && (23)\\
&= \frac{27}{6} && (24)\\
&= \frac{9}{2} && (25)
\end{aligned}
$$

Task 4

The sum of the first ten terms of an arithmetic sequence is 90.

Determine the common difference d, given that $a_1 = 3$.

Suggested solution

The sum of the first n terms of an arithmetic sequence is

$$S_n = \frac{n}{2}(2a_1 + (n-1)d)$$

Here, $S_{10} = 90$, $a_1 = 3$, $n = 10$.
Substitute into the formula:

$$
\begin{aligned}
90 &= \frac{10}{2}(2 \cdot 3 + 9d) && (26)\\
90 &= 5(6 + 9d) && (27)\\
90 &= 30 + 45d && (28)\\
60 &= 45d && (29)\\
d &= \frac{60}{45} = \frac{4}{3} && (30)
\end{aligned}
$$

Task 5

The points $A(1, 2, 0)$, $B(4, 6, 2)$, and $C(8, 6, 4)$ are given.

a) Determine the area of $\triangle ABC$.

Suggested solution

Find the vectors:

$$\vec{AB} = [3, 4, 2], \qquad \vec{AC} = [7, 4, 4]$$

The area of $\triangle ABC = \frac{1}{2}|\vec{AB} \times \vec{AC}|$:

$$\begin{aligned} \vec{AB} \times \vec{AC} &= \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ 3 & 4 & 2 \\ 7 & 4 & 4 \end{vmatrix} && (31) \\ &= [4 \cdot 4 - 2 \cdot 4,\ 2 \cdot 7 - 3 \cdot 4,\ 3 \cdot 4 - 4 \cdot 7] && (32) \\ &= [8,\ 2,\ -16] && (33) \end{aligned}$$

$$\begin{aligned} |\vec{AB} \times \vec{AC}| &= \sqrt{8^2 + 2^2 + 16^2} && (34) \\ &= \sqrt{324} = 18 && (35) \end{aligned}$$

$$\text{Area} = \tfrac{1}{2} \cdot 18 = 9$$

b) Find the distance from point C to the line through A and B.

Suggested solution

Use the relationship between area, base length, and height:

$$\begin{aligned} \text{Area} &= \tfrac{1}{2} \cdot |\vec{AB}| \cdot d && (36) \\ d &= \frac{2 \cdot \text{Area}}{|\vec{AB}|} && (37) \end{aligned}$$

$$|\vec{AB}| = \sqrt{3^2 + 4^2 + 2^2} = \sqrt{29}$$

$$d = \frac{18}{\sqrt{29}} = \frac{18\sqrt{29}}{29} \qquad (38)$$

Task 6

Given the planes $\alpha : x + y + z = 6$ and $\beta : 2x - y + z = 3$.

Find a parametric equation for the line of intersection between the two planes.

Suggested solution

The normal vectors of the planes are $\vec{n_\alpha} = [1, 1, 1]$ and $\vec{n_\beta} = [2, -1, 1]$.
The direction vector of the line is $\vec{r} = \vec{n_\alpha} \times \vec{n_\beta}$:

$$\begin{aligned} \vec{n_\alpha} \times \vec{n_\beta} &= \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} && (39) \\ &= [1 \cdot 1 - 1 \cdot (-1),\ 1 \cdot 2 - 1 \cdot 1,\ 1 \cdot (-1) - 1 \cdot 2] && (40) \\ &= [2,\ 1,\ -3] && (41) \end{aligned}$$

Find one point on the line by setting $z = 0$:

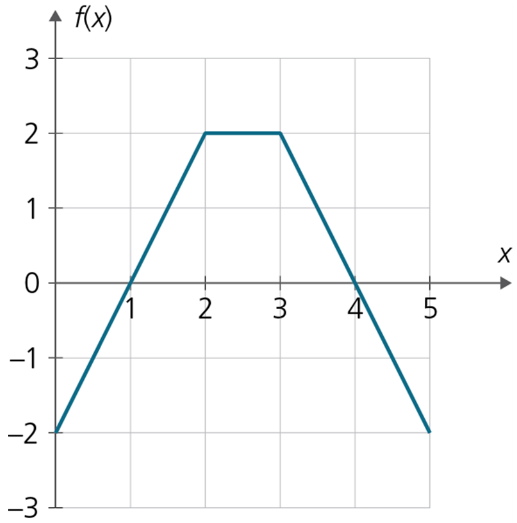
$$x + y = 6 \tag{42}$$
$$2x - y = 3 \tag{43}$$

Add the equations: $3x = 9 \implies x = 3,\ y = 3$.
Parametric equation of the line of intersection:

$$\ell\colon [3,\ 3,\ 0] + t[2,\ 1,\ -3], \quad t \in \mathbb{R}$$

Task 7

The figure below shows the graph of the function f.



The function g is given by $g(x) = \int_0^x f(t)\,dt$.

Determine $g(4)$ and $g'(4)$.

Suggested solution

$$g(4) = \frac{1}{2} \cdot 1 \cdot (-2) + \frac{(1+4) \cdot 2}{2} \tag{44}$$
$$= -1 + 5 \tag{45}$$
$$= 4 \tag{46}$$

Use the Fundamental Theorem of Calculus to find $g'(4)$.

$$g'(4) = f(4) \tag{47}$$
$$= 0 \tag{48}$$

Task 8

An angle θ in the second quadrant is given by $\sin\theta = \dfrac{3}{5}$.

a) Determine $\cos\theta$ and $\tan\theta$.

Suggested solution

Use $\sin^2\theta + \cos^2\theta = 1$:

$$\cos^2\theta = 1 - \sin^2\theta = 1 - \frac{9}{25} = \frac{16}{25} \tag{49}$$

In the second quadrant, $\cos\theta < 0$, so $\cos\theta = -\frac{4}{5}$.

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{-4/5} = -\frac{3}{4}$$

b) Solve the equation $2\cos(2x) = 1$ for $x \in [0, 2\pi]$.

Suggested solution

$2\cos(2x) = 1 \implies \cos(2x) = \frac{1}{2}$

Let $u = 2x$, then $u \in [0, 4\pi]$ and we solve $\cos u = \frac{1}{2}$:

$$u = \frac{\pi}{3} + 2k\pi \quad \text{or} \quad u = -\frac{\pi}{3} + 2k\pi, \quad k \in \mathbb{Z} \tag{50}$$

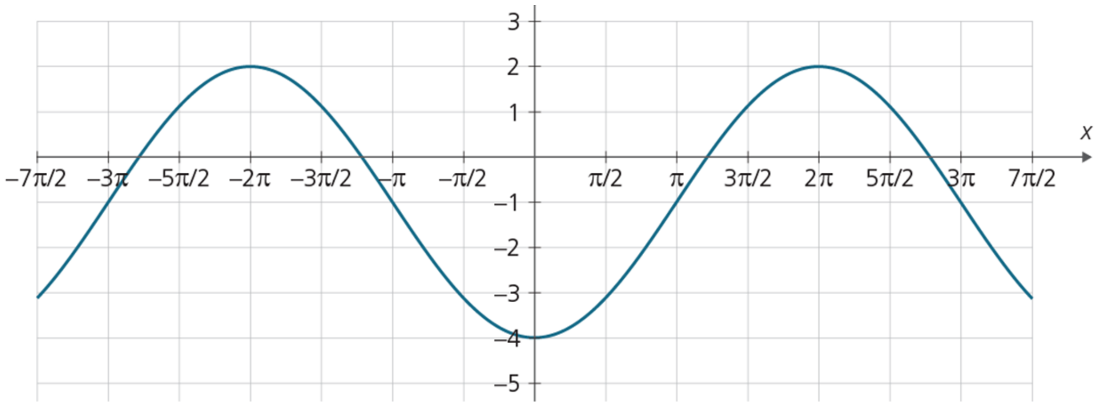
For $u \in [0, 4\pi]$:

$$u = \frac{\pi}{3}, \quad \frac{5\pi}{3}, \quad \frac{7\pi}{3}, \quad \frac{11\pi}{3}$$

Back to $x = \frac{u}{2}$:

$$x = \frac{\pi}{6}, \quad \frac{5\pi}{6}, \quad \frac{7\pi}{6}, \quad \frac{11\pi}{6}$$

The figure below shows the graph of a function f.



c) Find an expression for f.

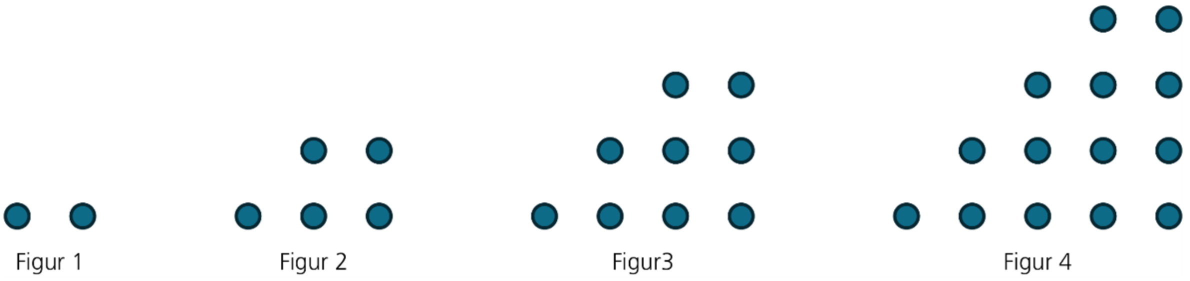
Suggested solution

Period: $p = 4\pi$.
Amplitude: $A = 3$.
Midline: $d = -1$.
Angular frequency: $c = \frac{2\pi}{p} = \frac{2\pi}{4\pi} = \frac{1}{2}$
To find the phase angle, first observe where the graph crosses the midline while increasing. This happens, for example, at $x = x_0 = \pi$.
Then $cx_0 + \phi = 0$, which gives $\phi = -cx_0 = -\frac{\pi}{2}$.

> The harmonic oscillation is:
>
> $$f(x) = 3\sin\left(\frac{1}{2}x - \frac{\pi}{2}\right) - 1$$

Task 9

Each figure below consists of dots arranged in a specific pattern.



Let F_n be the number of dots in figure n. The first four figure numbers are 2, 5, 9, and 14.

a) Determine F_6.

Suggested solution

$F_6 = 27$

b) It can be shown that $F_n = F_{n-1} + n + 1$. Carry out an induction proof showing that $F_n = \frac{1}{2}n^2 + \frac{3}{2}n$ for all $n \geq 1$.

Suggested solution

We will show that $F_n = \frac{1}{2}n^2 + \frac{3}{2}n$ for all $n \geq 1$.
Base case ($n = 1$):

$$F_1 = \frac{1}{2} \cdot 1^2 + \frac{3}{2} \cdot 1 = \frac{1}{2} + \frac{3}{2} = 2\checkmark$$

Induction step: Assume that $F_k = \frac{1}{2}k^2 + \frac{3}{2}k$ for an arbitrary $k \geq 1$ (the induction hypothesis).
We must show that $F_{k+1} = \frac{1}{2}(k+1)^2 + \frac{3}{2}(k+1)$.
From the recursive relation $F_n = F_{n-1} + n + 1$, we get:

$$F_{k+1} = F_k + (k+1) + 1 = F_k + k + 2$$

Substitute the induction hypothesis:

$$F_{k+1} = \frac{1}{2}k^2 + \frac{3}{2}k + k + 2 \tag{51}$$
$$= \frac{1}{2}k^2 + \frac{5}{2}k + 2 \tag{52}$$

Check that the formula gives the same value for $n = k + 1$:

$$\frac{1}{2}(k+1)^2 + \frac{3}{2}(k+1) = \frac{1}{2}(k^2 + 2k + 1) + \frac{3}{2}k + \frac{3}{2} \tag{53}$$
$$= \frac{1}{2}k^2 + k + \frac{1}{2} + \frac{3}{2}k + \frac{3}{2} \tag{54}$$
$$= \frac{1}{2}k^2 + \frac{5}{2}k + 2\checkmark \tag{55}$$

The formula is proven for all $n \geq 1$.

Task 10

Aisha is saving for a down payment to buy a home. On the 1st of each month, she transfers NOK 12,000 to a savings account with 0.3% monthly interest. The first transfer was on 1 January 2023.

a) Explain why the very first deposit has been in the account for 36 months by the end of December 2025.

Suggested solution

From 1 January 2023 to 31 December 2025 is exactly 3 years $= 3 \times 12 = 36$ months. The first deposit was made on 1 January 2023 and has therefore been in the account for 36 months by the end of December 2025.

b) How much does she have in the account by the end of December 2025?

Suggested solution

Deposit number j (counted from the end) has been in the account for j months and has grown to $12\,000 \cdot 1{,}003^j$.
The sum of all 36 deposits is a geometric series with $a_1 = 12\,000 \cdot 1{,}003$, ratio $k = 1{,}003$, and $n = 36$ terms:

$$
\begin{aligned}
S &= 12\,000 \cdot 1{,}003 + 12\,000 \cdot 1{,}003^2 + \cdots + 12\,000 \cdot 1{,}003^{36} && (56)\\
&= \frac{12\,000 \cdot 1{,}003 \cdot (1{,}003^{36} - 1)}{1{,}003 - 1} && (57)\\
&= \frac{12\,000 \cdot 1{,}003 \cdot (1{,}003^{36} - 1)}{0{,}003} && (58)\\
&\approx 456\,836{,}99 \text{ NOK} && (59)
\end{aligned}
$$

c) How much would she need to deposit each month for the account balance to be half a million NOK by the end of December 2025?

Suggested solution

Let x be the monthly deposit. Then:

$$\frac{x \cdot 1{,}003 \cdot (1{,}003^{36} - 1)}{0{,}003} = 500\,000$$

Solve for x:

$$
\begin{aligned}
x &= \frac{500\,000 \cdot 0{,}003}{1{,}003 \cdot (1{,}003^{36} - 1)} && (60)\\
&\approx 13\,133{,}79 \text{ NOK} && (61)
\end{aligned}
$$